Pavan Kumar Gannoju

Backend and Systems Engineer · Hyderabad, India · open to relocate to Germany
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PROFILE

Backend and systems engineer (Java, Go, Rust, Python) who checks correctness with fault injection, property-based tests, fuzzing and CI. Built a double-entry ledger with benchmarked concurrency strategies, a Raft-replicated key-value store validated with a linearizability checker, and a capability-based OS kernel whose revocation is asserted in CI. Every figure comes from a committed, re-runnable measurement.

WORK EXPERIENCE

03/2025 – 11/2025
Remote, part-time

AI Intern → Research Team Lead, Prodigal AI Technologies Pvt. Ltd., Delhi

- Promoted from AI Intern to Research Team Lead; coordinated a small engineering team through architecture reviews and sprint delivery for internal AI workflow systems.
- Built modular Python backend workflows and API integration utilities for data processing and cross-service coordination; owned request validation and issue resolution end to end.

05/2025 – 09/2025
Remote, part-time

Gen AI / LLM Intern, Digital Nexus AI, Jersey City, NJ, USA

- Developed 5+ backend REST APIs in Python and FastAPI supporting inter-service communication and automated workflow execution.
- Supported deployment workflows in Linux cloud environments; contributed to integration testing and defect triage ahead of release cycles.

PROJECTS

2026
Java 21, Spring Boot, PostgreSQL, Kafka, Testcontainers, k6

LedgerLine — Double-Entry Ledger (GitHub)

- Checked integer-money transfers with PostgreSQL zero-sum constraints, non-negative customer balances, principal-scoped idempotency, a transactional outbox and deduplicating Kafka projections.
- Temporary Azure validation: 5,235 successful transfers, zero HTTP failures, and 10,484 ledger postings matching projected statements with zero statement or daily-summary drift; resources deleted after validation.
- 53 Java tests and 4 image-scan policy tests passed in CI; regressions cover late events, overflow rollback and replay. Walkthrough · Evidence. Portfolio demonstration, not production banking.

2026
Go, gRPC, hashicorp/raft, Porcupine

QuorumKV — Replicated Key-Value Store (GitHub)

- CRC-framed, fsync-before-ack write-ahead log, proven with 100/100 randomized process-kill trials: no acknowledged write lost, none invented.
- Fault-injection harness (TCP proxy per directed link) over a 3-node Raft cluster; recorded histories for 5 scenarios (leader and minority partitions, rolling restarts, network delay, double-leader attempt) checked with Porcupine; the checker was validated against known-bad histories (12 tests), and errored writes are modelled as "may have happened".

2026
Python, FastAPI, PyTorch, Redis, Docker

Gatekeeper — Fail-Closed LLM Guardrail Gateway (solo; GitHub)

- Prompt-injection and PII gateway (symbolic rules, FAISS search, learned fusion of 8 detectors, server-side privilege resolution, audit log); 1,644 tests, CI with dependency audit and Trivy/SBOM image scanning.
- Profiled with py-spy, found CPU oversubscription under concurrency, and caught a cache-warmth flaw in my own first benchmark; pinning torch threads raised throughput from about 10.3 to 16.0 rps and cut p95 from about 2.1 s to 1.3 s at 16 concurrent requests (same workload, one machine; p99 unchanged).
- Found and fixed an authorization bypass: a client-supplied *role* field could set INTERNAL and disable every guardrail; capability is now resolved server-side and the field is rejected with a 422.

2026
Rust (no_std), UEFI, QEMU, GitHub Actions

Agentic-OS — Capability-Based x86_64 UEFI Kernel (solo; GitHub)

- All access goes through per-process capabilities: rights attenuation enforced on derive, O(1) generation-counter revocation, drivers in user space.
- Linux CI builds the kernel, runs host tests, boots it in QEMU and asserts capability revocation; ext2 parsers are fuzzed. Runs under QEMU only.
- Traced a non-deterministic CI-only kernel page fault with added instrumentation; fixed a missing LAPIC end-of-interrupt in the timer handler.

2026
Python, FastAPI, Ollama, Docker

Enterprise RAG — FP&A Backend Platform (GitHub)

- Bounded concurrency cut a 181-chunk ingest from 378.7 s to 48.1 s (7.83×); a 1,408-chunk corpus ingests in 87 s. Scale exposed a real defect (a connection reset raised *OSError*, not *URLError*, and was never retried), fixed with a regression test.
- Load test separated rate-limiter policy (29/40 served, 11 rejected with 429) from capacity (40/40 at a raised limit, about 1.5 req/s); 343 tests, 98% branch coverage.

EDUCATION

to 04/2026 — **B.Tech, Computer Science & Engineering (AI/ML)**, Visvesvaraya College of Engineering and Technology (JNTUH) · CGPA 8.01/10.0

2024 – 2026 — **IIT Ropar** — Major in Artificial Intelligence (online programme via Masai/NSDC)

SKILLS

Languages	Java, Go, Rust, Python, C/C++, SQL, JavaScript
Backend and infrastructure	Spring Boot, FastAPI, gRPC, Kafka, REST APIs, Docker, GitHub Actions, AWS, Linux
Databases	PostgreSQL, MySQL, SQLite
Testing and verification	Testcontainers, property-based testing, fuzzing, Porcupine, k6
Core CS	Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems

LANGUAGES

English	IELTS 6.5 (overall band)

CERTIFICATIONS AND FURTHER ACTIVITIES

Certificates	Google Cybersecurity Professional Certificate · Microsoft Cybersecurity & OS Fundamentals · SAP Code Unnati Advanced Training Program · Google Data Analytics
Competitive programming	LeetCode 530+ solved including 86 Hard · Codeforces 249 problems solved (218 in 2026)
Leadership	Led a 5-member development team on backend implementation and code reviews · Organised and mentored two university hackathons (50+ participants)